Diagnosing the Problem Itself: Behavioural Signals in the Sprintlab LIS

A psychological-perspective literature review of Framework Sections 4–6

Question of record: "How do we find the problems themselves (diagnosis), and what are the signals?" • Scope: the 12 non-content constructs in §4 Cognitive Skills, §5 Exam Skills, §6 Psychological & Behavioural Patterns • XES3G5M used as proxy only.

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1. The question, and why a literature review

Two students can give the *identical* wrong answer for different reasons — never learned it, knew it but rushed, misread it, or hold a confident misconception. Accuracy alone cannot separate these; only *behavioural* signals (response time, error *pattern*, response to game events) can. The Master Framework already asserts one detection rule per construct, but flags every rule as an unvalidated hypothesis (§9–§10). This review therefore asks the prior, reviewer's question for each of the 12 constructs: **is the asserted signal grounded in the psychology / cognitive-science / learning-analytics literature, and can it actually be measured?** Each construct receives a verdict — **Supported**, **Partial**, or **Weak** — the seminal grounding, and the dominant confound the literature warns about.

Sub-questions (per construct). SQ1 What is the construct and how does psychology measure it? **SQ2** Is the Framework's proposed telemetry signal valid? **SQ3** What confound must be controlled to keep it separable from plain not-knowing? **Cross-cutting: SQ4** trait vs. state; **SQ5** what does our data actually permit?

2. Findings I — Cognitive Skills (§4): raw processing capacity

Construct	Framework signal	Verdict	Seminal grounding	Dominant confound & the fix
Working Memory	Collapses on high-load items (load 4–5) but not low-load, same cluster	**Partial**	Baddeley & Hitch 1974; Cowan 2001 (≈4 chunks); Sweller 1988 (element interactivity)	High-load items are also *harder* and more knowledge-demanding; expert schemas chunk load away. *Hold content mastery constant across load levels.*
Processing Speed	Accurate but RT near the max even on items answered correctly	**Weak**	Salthouse 1996; Kail 1991	Framework also predicts "errors rise" — that is a speed–accuracy trade-off, not pure slowness. *Clean signal = slow **and correct** on an easy, already-mastered item.*
Selective / Sustained Attention	Errors when a distractor tile is present (selective); accuracy decays over the last items (sustained)	**Supported** **Weak**	Posner & Petersen 1990; Eriksen & Eriksen 1974 (flanker); Mackworth 1948; Robertson 1997 (SART)	Selective is a valid flanker analogue *if the distractor is truly irrelevant*. Vigilance decrement needs 20+ min of monotony — a 10-item match instead shows rushing / difficulty-order. *De-confound position from difficulty.*
Cognitive Flexibility	Accuracy/RT drop after a *subject* switch (chemistry→physics)	**Weak**	Miyake et al. 2000 (shifting); Rogers & Monsell 1995 (switch cost); Diamond 2013	In the literature the "task set" is the S–R rule, not the academic subject; a subject change likely measures knowledge-set *restart cost*, not executive shifting. *Vary the required operation, hold subject constant.*

3. Findings II — Exam Skills (§5): tactical test-taking

Construct	Framework signal	Verdict	Seminal grounding	Dominant confound & the fix
Question Interpretation & Reading	Picks the reading-trap distractor at *fast*-to-average RT	**Partial**	Millman, Bishop & Ebel 1965 (test-wiseness); Messick 1989 (construct-irrelevant variance); Clark & Chase 1972 (negation cost)	Fast + trap is equally the signature of rapid guessing / low effort. *Isolate reading via the same student's plain-vs.-language-challenging gap on matched content.*
Time Management & Perfectionism	(a) high timeout rate; (b) 90% of time on an easy item	(a) **Supported** (b) **Weak**	Wise & Kong 2005 (RT effort); Schnipke & Scrams 1997 (speededness); Frost et al. 1990 (perfectionism)	Timeout also = difficulty / slow speed. "Easy" is population-level, not per-student. *Norm dwell to the student; slow-**correct** (checking) vs. slow-**wrong** (difficulty) is the disambiguator.*
Reasoning	Fails short high-logical-step items but solves data-heavy low-step ones	**Partial**	Carroll 1993 / Cattell–Horn Gf; Halford et al. 1998 (relational complexity); Frederick 2005 (CRT)	Relational complexity is *defined* via processing capacity, so it is entangled with working memory. *RT splits it: engaged-but-wrong (slow) = Gf ceiling; fast-wrong = reflection failure.*

4. Findings III — Psychological & Behavioural Patterns (§6): pressure responses

Construct	Framework signal	Verdict	Seminal grounding	Dominant confound & the fix
Impulsiveness	Picks the intuitive trick-lure in <30% of theoretical time	**Supported**	Kagan 1966 (reflection–impulsivity, MFFT); Frederick 2005 (CRT lure); Patton, Stanford & Barratt 1995 (BIS-11)	Kagan's own signature is exactly short-latency × error. Fast-wrong also fits low effort. *Require the error to be the* ***specific lure****; 30% is arbitrary — normalise to the person.*
Stress & Overthinking	After a game trigger: wrong easy answer, or RT >150% of own norm	**Partial**	Liebert & Morris 1967 (worry vs. emotionality); Cassady & Johnson 2002; Eysenck et al. 2007 (Attentional Control Theory)	Event-gating is correct (state anxiety is trigger-bound); RT-inflation matches ACT's efficiency cost, but also indexes careful reflection. *Trigger + RT-inflation + accuracy* ***preserved*** *= worry, not tilt.*
Low Self-Efficacy	Switches final answer *away* from an initially-correct one	**Weak**	Bandura 1997; Benjamin et al. 1984; Kruger, Wirtz & Miller 2005	The base rate of answer-changes is wrong→right (adaptive); the correct→wrong tail is rare and noisy. *Need a repeated pattern, strict "no new information," and fast/uncertain reversals — never a one-shot flag.*
Low Effort / Motivation	Flat, fast RT regardless of difficulty; disengaged from optional mechanics	**Supported**	Wise & Kong 2005 (response-time effort); Baker et al. 2004 (gaming / off-task)	Genuine effort makes RT *scale* with difficulty, so flat-fast is a strong marker — but flat-fast + *high* accuracy is mastery. *Require low/chance accuracy before flagging.*
Low Frustration Tolerance (Tilt)	Post-negative-event performance drops and does not recover in-match	**Partial**	D'Mello & Graesser 2012 (affect dynamics); Rabbitt 1966 (post-error slowing); *esports tilt*, CHI 2021	A single post-event dip is *adaptive* (post-error slowing improves the next answer). *Only sustained non-recovery over a multi-item window is tilt; separate RT-up/acc-up from acc-down.*

5. Cross-cutting findings

SQ3 — one confound dominates all twelve. Response time and accuracy are jointly driven by *whether the student knows the material*: a student who simply doesn't know the answer looks slow, error-prone, distractible, "inflexible," and "stressed" at once. The literature neutralises this four ways, and all four should be design constraints for the LIS: (1) **within-student contrasts** — each student is their own baseline (this is exactly the Framework's Personal Pace Factor, §7.1); (2) **hold content & difficulty constant** while varying only the cognitive/tactical demand; (3) **condition on correctness** (analyse RT on correct responses only) and calibrate difficulty via IRT; (4) **jointly model speed & ability** (van der Linden 2007). This is the psychometric backing for the Framework's Trait-vs-State test and Funnel (§7.3).

SQ4 — trait vs. state sets the required evidence window. *Impulsiveness* and *Low Self-Efficacy* are stable **traits**: reliable only when accumulated across many items/sessions — matching the Framework's *Confirmed* tier (100+ Qs), and a single event is weak evidence. *Stress* and *Tilt* are transient **states**, validly read from tightly time-locked pre/post-trigger contrasts — so the Framework's event-gating is methodologically correct. *Low Effort* is mixed (a match-level state that becomes dispositional if chronic). Reading a trait from one state-like event is the framework's central measurement risk.

SQ5 — the proxy data cannot see most of these signals (evidence: companion notebook). Nearly every §4–6 signal is timing-based, yet XES3G5M has **no per-item latency**: 78.7% of responses share a timestamp with another and 58% of consecutive answers are $\Delta t = 0$ — timestamps are session/batch-level. Consequently only **2 of 12** constructs are even weakly observable on the proxy (sustained attention: a small monotone $0.825 \rightarrow 0.812$ accuracy decay over 10 items; a weak cognitive-load→difficulty trend). The subject-switch penalty is ≈0 because the proxy is single-domain math; distractor/trick and game-event signals need tags and event logs that do not exist in the logs.

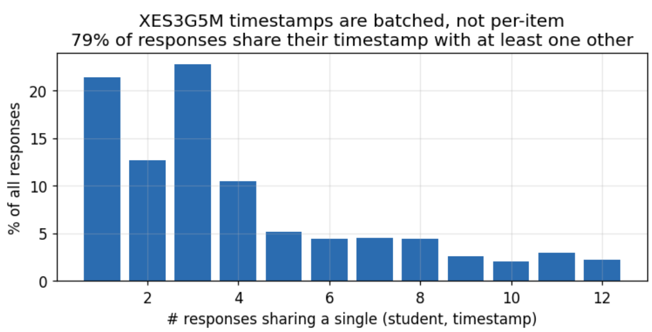
6. Recommendation — what Sprintlab must instrument

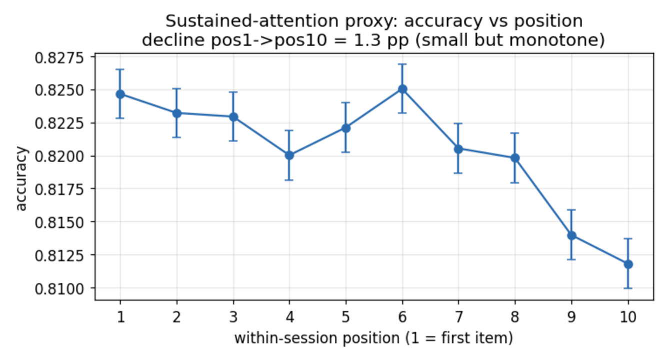
For 10 of 12 constructs the blocker is missing *data*, not diagnostic logic. Ranked by constructs unlocked: **(1) per-item response latency** (unlocks Processing Speed, Time Management, Low Effort, Impulsiveness, Stress, Reading); **(2) chosen-option + offline LLM distractor tags** (Impulsiveness, Reading traps, Misconception attribution); **(3) in-game event log** — triggers, attacks, distractor tiles (Selective Attention, Stress, Tilt); **(4) answer-change events** (Low Self-Efficacy); **(5) Variables_Count & Logical_Steps tags** (WM vs. Reasoning split); **(6) multi-subject interleaving** (Cognitive Flexibility). Every timing threshold (the 30%, 90%, 150% rules) must then be re-expressed as a within-student, per-load, grade-band-calibrated contrast before any diagnosis is trusted — none is defensible as a raw population cutoff.

7. Findings IV — can our data even see these signals? (companion notebook)

The diagnosis question has a data half: a signal we cannot compute is not a diagnosis. The audit below runs every §4–6 signal against the real proxy (XES3G5M: 5.14 M responses, 14 453 students). **Only 2 of 12 constructs are observable**, and the gap is almost always *missing telemetry*, not missing logic.

Construct	What the signal fundamentally requires	Verdict on XES3G5M
Sustained Attention	within-session position × accuracy	**COMPUTABLE** *(shown: 0.825→0.812)*
Working Memory	per-item load index × accuracy	**PROXY-ONLY** *(weak #KC trend)*
Reasoning	Logical_Steps tag × accuracy	**PROXY-ONLY**
Processing Speed	per-item latency + time-pressure tag	**NEEDS LATENCY**
Time Management	per-item latency + timeout events	**NEEDS LATENCY**
Low Effort / Motivation	per-item latency + engagement events	**NEEDS LATENCY**
Impulsiveness	trick/distractor tags + latency	**NEEDS TAGS**
Question Interpretation	reading-trap distractor tags + latency	**NEEDS TAGS**
Selective Attention	in-game distractor-tile events	**NEEDS GAME EVENTS**
Stress & Overthinking	game-state triggers + latency	**NEEDS GAME EVENTS**
Low Self-Efficacy	answer-change (initial vs. final) events	**NEEDS GAME EVENTS**
Cognitive Flexibility	multi-subject interleaving in the stream	**NEEDS MULTI-SUBJECT** *(penalty ≈0 on math-only proxy)*





Left — the fatal gap. XES3G5M `timestamps` are batched, not per-item: 78.7% of responses share their timestamp with another and 58% of consecutive answers have $\Delta t = 0$. No trustworthy per-item solving time can be recovered, which is why every timing-based signal (the majority of §4–6) is un-testable here. **Right — the one clean proxy signal.** Accuracy falls monotonically across a session (a sustained-attention analogue), but the effect is small (1.3 pp) because the proxy is untimed and non-competitive; the true in-game decrement is expected to be larger and must be measured on target data.

Implication for the build. The proxy is strong for *content* diagnosis (clean learning and differentiation curves) and structurally silent on *behavioural* diagnosis — exactly the §4–6 target. Behavioural signals therefore cannot be validated on borrowed knowledge-tracing data; they require Sprintlab's own instrumentation (§6 above), after which the thresholds must be re-calibrated per student and grade band before any label is trusted. *Full reproduction:* `notebooks/signal_feasibility_audit.ipynb`.

Key sources. Baddeley & Hitch (1974); Cowan (2001); Engle (2002); Sweller (1988); Salthouse (1996); Kail (1991); Posner & Petersen (1990); Eriksen & Eriksen (1974); Mackworth (1948); Robertson et al. (1997); Miyake et al. (2000); Rogers & Monsell (1995); Diamond (2013); Millman, Bishop & Ebel (1965); Messick (1989); Clark & Chase (1972); Wise & Kong (2005); Schnipke & Scrams (1997); Frost et al. (1990); Carroll (1993); Halford et al. (1998); Frederick (2005); Kagan (1966); Patton, Stanford & Barratt (1995); Liebert & Morris (1967); Cassady & Johnson (2002); Eysenck et al. (2007); Bandura (1997); Benjamin et al. (1984); Kruger, Wirtz & Miller (2005); Baker et al. (2004); D'Mello & Graesser (2012); Rabbitt (1966); van der Linden (2007). *Citations verified for author/year/venue; DOIs and effect sizes deliberately not asserted.*